

KLM SCROD SRAM Expansion

Feb. 2026 Update



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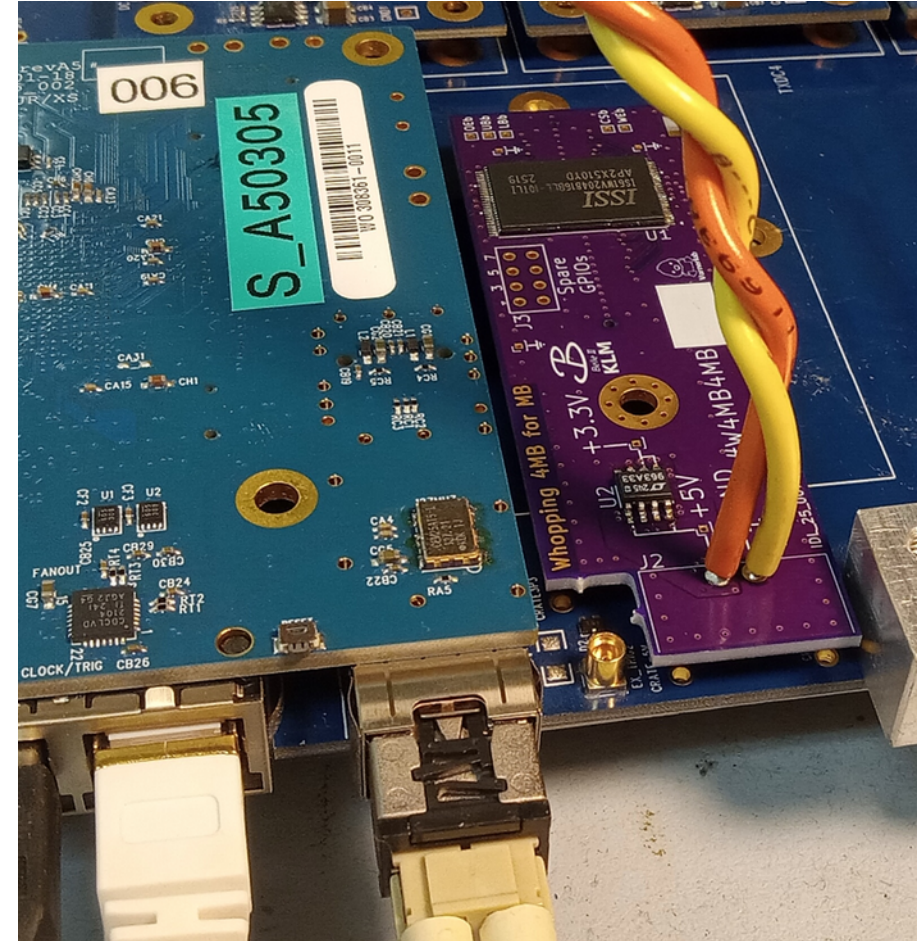


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- ☕ In events with multi-strip hits, SRAM access for fetching stored pedestals can become the bottleneck in waveform processing
- ☕ Scintillator motherboard has spare 80 pin connector with nets connected to FPGA GPIOs
- ☕ Got approval to create SRAM expansion PCB
 - ☕ Now each axis (called Bus A / Bus B on motherboard) will have its own SRAM chip for pedestal storage
 - ☕ New scheme will halve the time for pedestal access

SRAM UPDATE

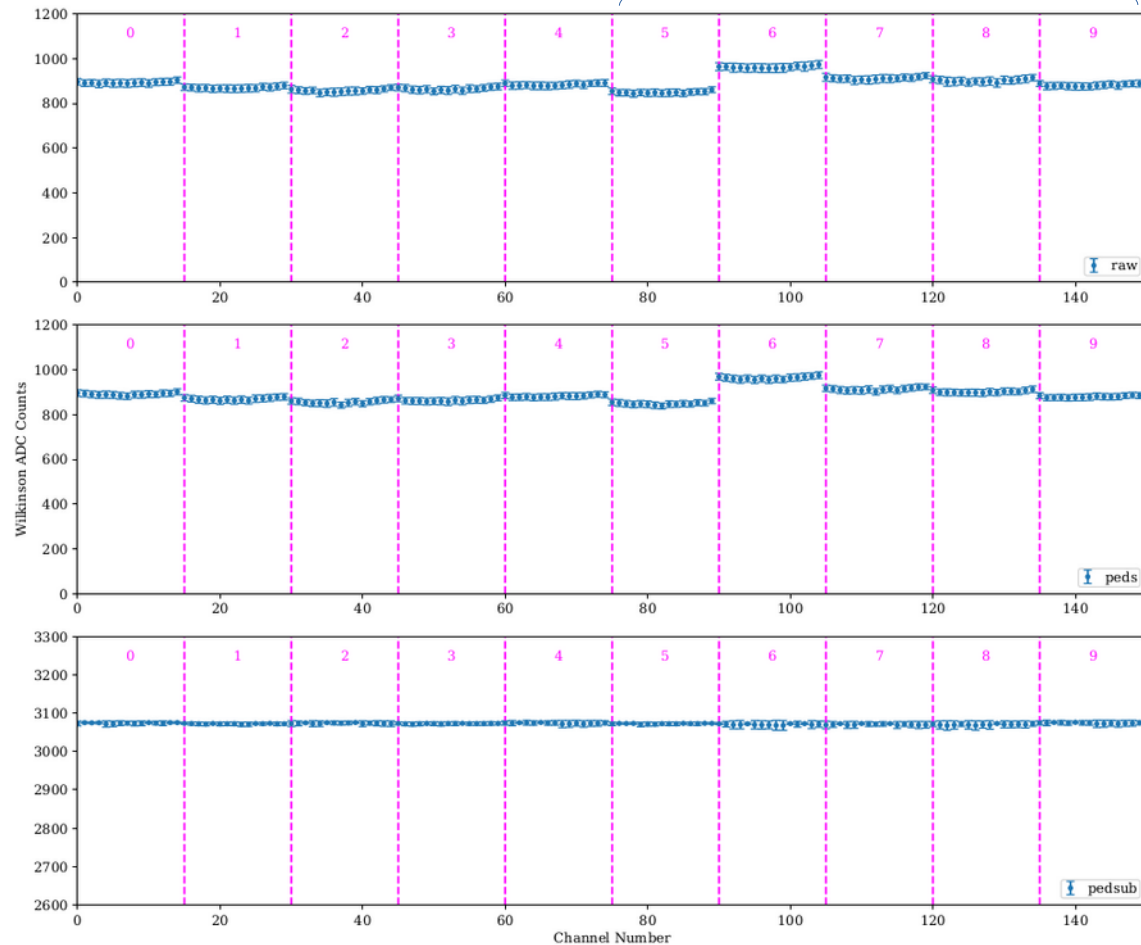
- ☕ Prototype board designed
- ☕ 1st article soldered together
 - ☕ Still finalizing choice for 5V connector
- ☕ New firmware prepared last summer by Horacio and I
- ☕ After modifying the pin mapping in firmware, our modified SRAM access logic in the SCROD firmware worked great



TEST RESULTS (128 KSAMPLES)

Results are consistent with similar scan using only the original SRAM chip for pedestal storage!

Bus B (New SRAM)



VERSION LOG

☕ v3201

- ☕ First compile of SRAM upgrade firmware

☕ v3202

- ☕ Tied **BUS_REGCLR** to GND as in master branch

☕ v3203

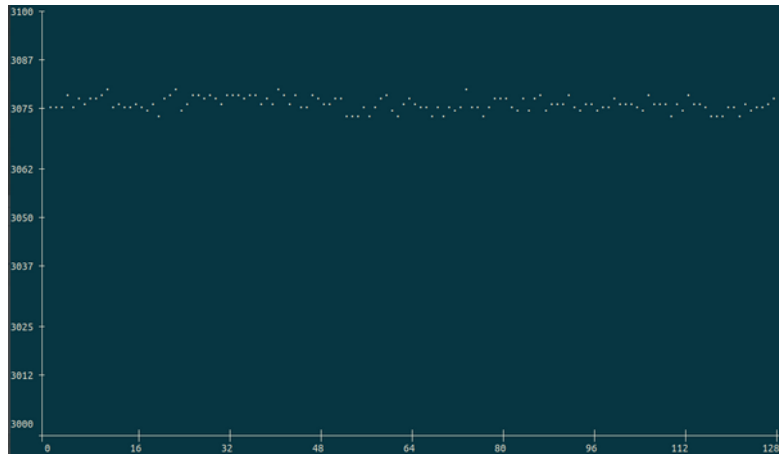
- ☕ Fix pin mapping in UCF file

☕ v3301

- ☕ Merge with vFE33 and **feature/ramp_settle**

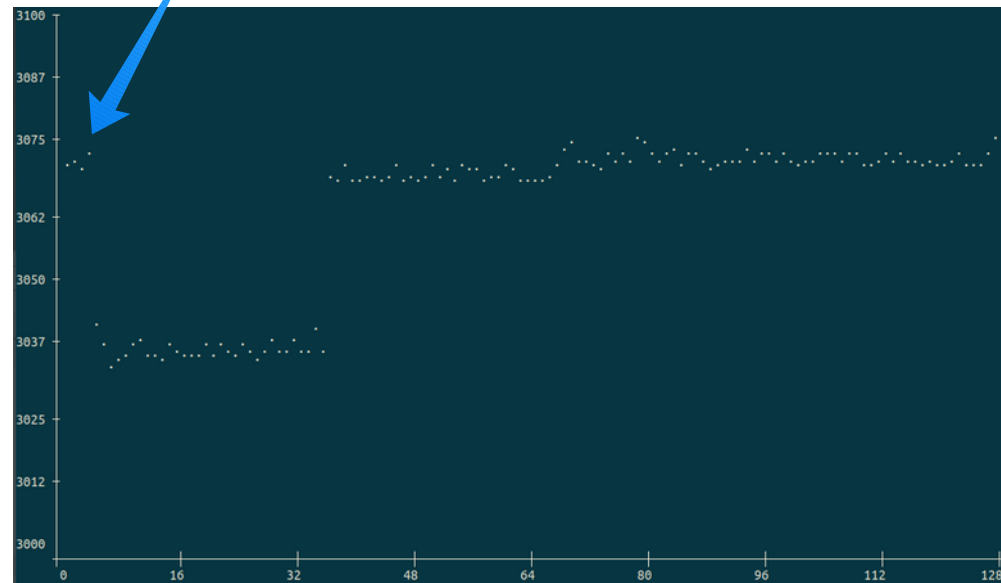
What's that?

ramp_settle EXPLAINED



Normal

First window only
shifts last 4 samples



Ramp reactivates too soon --- higher starting
voltage ==> fewer ADC counts

VERSION LOG



v3201



First compile of SRAM upgrade firmware



v3202



Tied **BUS_REGCLR** to GND as in master branch



v3203



Fix pin mapping in UCF file



v3301



Merge with vFE33 and **feature/ramp_settle**

- **ramp_settle** feature not working as anticipated (Wilkinson counter starts too early)



v3401



Merge with vFE34



v3402



Refactor **feature/ramp_settle**
(delay **BUS_CLR** and **BUS_RD_ENA** along with **BUS_RAMP**)

Now 2nd-window ADC performance is restored!

... but

Latest bit file shows outliers in pedestals from new SRAM (hardware seems okay --- need to investigate further)

REMAINING TASKS



SRAM Expansion PCB

- Finalize +5V connector choice
(Cheap, easy, secure, non-interference)
- Double-check PCB design
(Characteristic impedance? Image-current path? Happy with silkscreen?)
- Order 3 more, solder & test, then order production run



Design RTM board for +5V supply



Design wire harness

(Determine gauge from power budget, appropriate flexibility or stiffness, how to sheath and preform bends)



Investigate feasibility of securing SRAM PCB with standoff

- Enough threads for proper engagement?
- Necessary screw sizes available to fully utilize threads?
- Screw heads don't interfere with neighbor boards?



Investigate SCROD fastening hardware at same time

- Same considerations as above
- Ease of installation?



Order all hardware, wire spools, and ICs



Design clamp tool for easy MB removal

TIMELINE

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Soldered 1st board											
	Testing										
		Finalize board designs									
			-----Production run----- (Send to board fab. house)								
			Build Wiring Harnesses in house (12 required)			-----Installation-----					

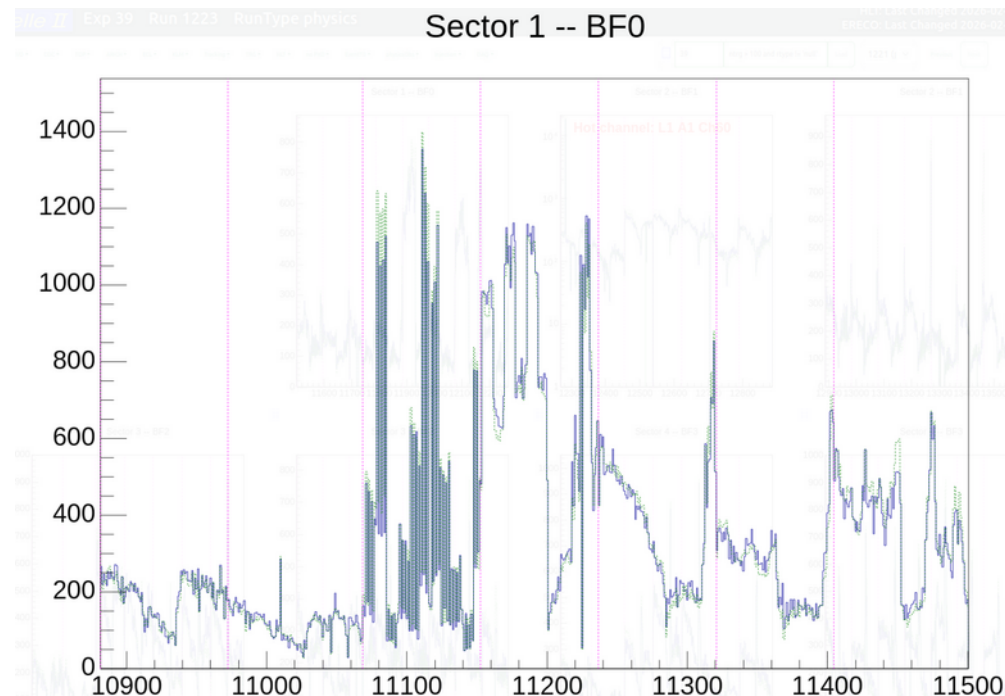
Backup

OPERATIONS

☕ On shift all week

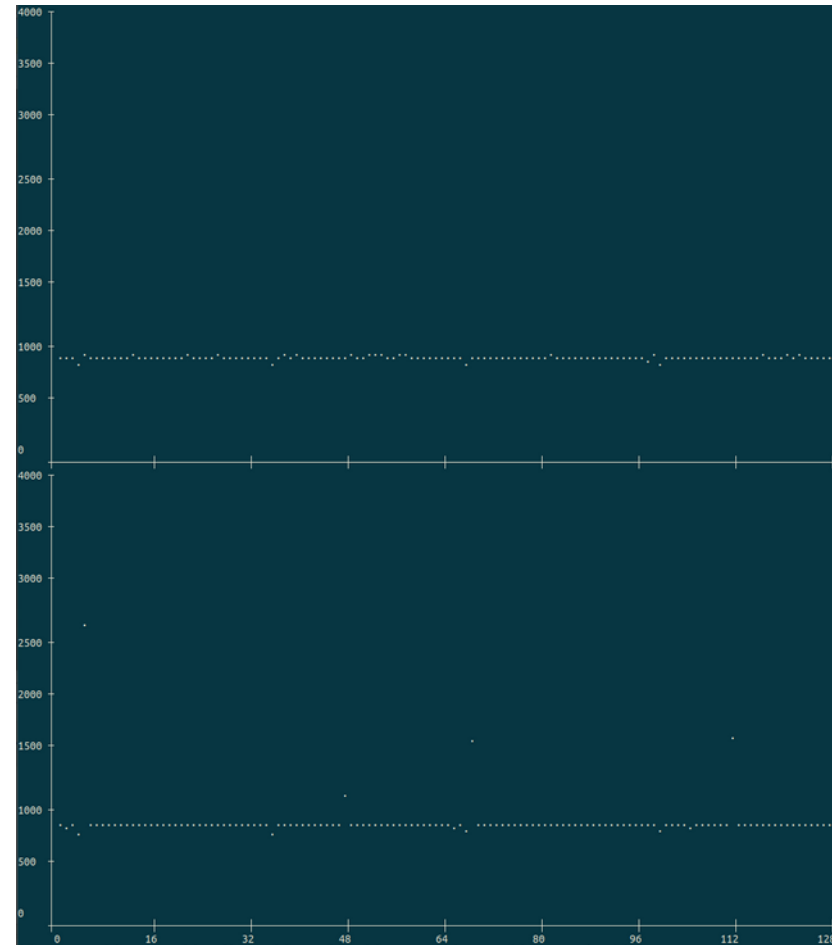
☕ Data Concentrator dc13

- ☕ Installed in BF0
- ☕ 3rd layer frequently drops in efficiency after certain beam aborts
 - Twice yesterday, the first saw 1300 mrad to beampipe diamonds
 - Recovers after reprogramming
 - Hit-channels distribution shows comb-like structure (as though every other channel has an issue)
 - Maybe we should replace the RPC front-end board ...
- ☕ BB6 3rd layer shows similar hit-channel distribution!



PEDESTAL OUTLIERS

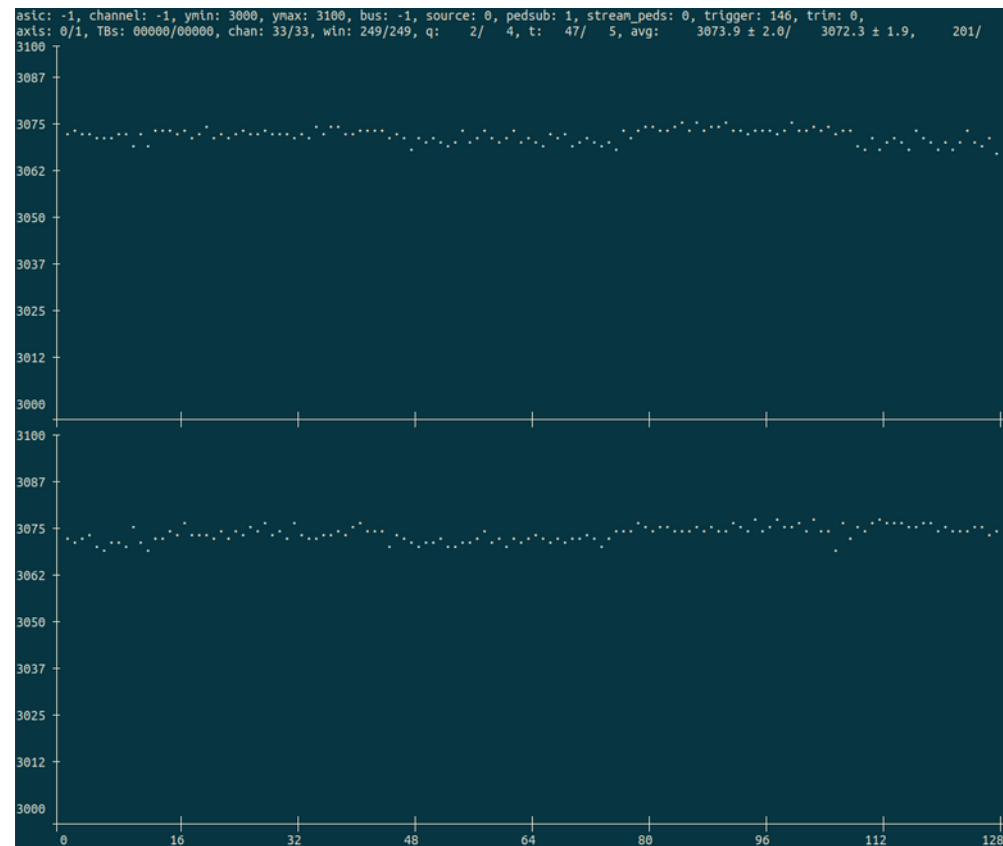
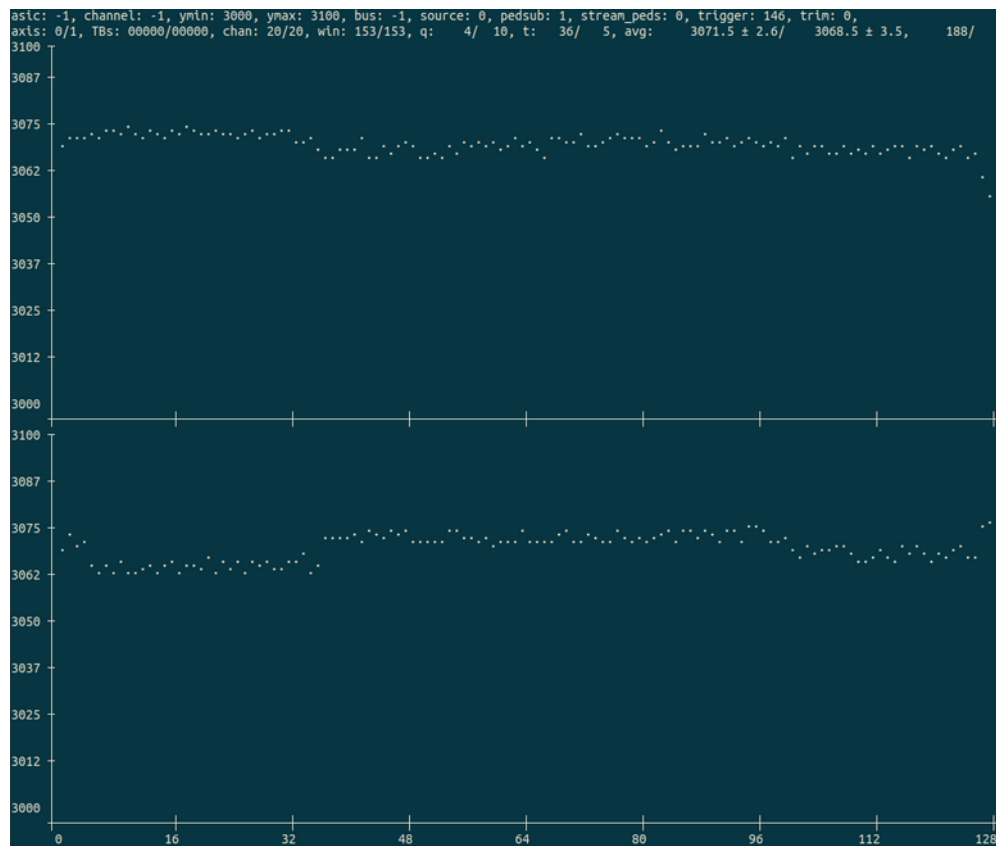
- ☕ Top: Bus A
Bottom: Bus B (new SRAM)
- ☕ Streaming pedestal values only (direct from SRAM chips to debug wave FIFO)
- ☕ Four outliers in this example waveform (typical)
- ☕ Reverting to previous bitfile fixed issue, so new hardware is not the issue
- ☕ Firmware logic changes completely unrelated
- ☕ Suspect cross-talk from unconstrained GPIO pins --- will investigate on oscilloscope





Compiled v3403

- Grounded spare GPIOs that are routed to SRAM expansion
- Outlier problem fixed
- ADC drop in 2nd digitized window seems much much better
 - Basically perfect most of the time, but a few examples of a slight dip in the 2nd window (~10 ADC counts as opposed to sometimes 40 or more seen previously)





Want to try adding pulldown resistors to
SR Clock and TARGETX DO



Can we shift out data faster?



Does it change the current draw?



